

Mess Management System (Phase 1) - Software Requirements Specification

Project Overview

Develop a modern, secure, responsive, and user-friendly web-based **Mess Management System** for managing boarding houses/student messes. The system should automate meal tracking, market management, expenses, payments, reports, and member management while supporting both **English and Bangla**.

The system must be designed with scalability in mind so future features (such as Mobile App, Online Payment Integration, and Multi-Mess Support) can be added without major architectural changes.

Technology Stack

Frontend

- React.js
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- Supabase PostgreSQL

Authentication

- Supabase Auth

Version Control

- GitHub

Hosting

- Frontend: Vercel (Free)
 - Backend: Render (Free)
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User Roles

Admin

Admin has complete control over the entire system.

Member

Members have limited access and can only perform operations assigned to them.

Authentication

- Admin Login
 - Member Login
 - Logout
 - Forgot Password
 - Change Password
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Dashboard

Display:

- Total Members
 - Active Members
 - Total Market Cost
 - Total Expenses
 - Total Meals
 - Current Meal Rate
 - Total Collection
 - Total Due
 - Today's Market
 - Recent Activities
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Member Management

Admin can:

- Add Member
- Edit Member

- Delete Member
- Activate Member
- Deactivate Member

Store:

- Full Name
 - Phone Number
 - Room Number
 - Joining Date
 - Profile Photo (Optional)
 - Status
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Meal Management

Meal Types

- Breakfast
- Lunch
- Dinner

Automatically calculate

- Daily Meal
 - Monthly Meal
 - Total Meal
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Smart Market Management

Predefined Shopping List

System should include commonly used items such as

- Rice
- Flour
- Lentils
- Salt
- Sugar
- Cooking Oil
- Onion
- Garlic
- Ginger
- Potato

- Tomato
- Green Chili
- Fish
- Chicken
- Beef
- Eggs
- Turmeric
- Chili Powder
- Coriander
- Cumin
- Soap
- Tissue
- Detergent
- Dishwashing Liquid
- Other daily essentials

Admin should be able to add or remove predefined items.

Members should simply select items instead of typing them repeatedly.

Market Entry

Fields

- Market Date
- Buyer
- Item
- Quantity
- Unit
- Price
- Notes
- Receipt Image (Optional)

Support custom items.

Market Management Permissions (Role-Based Access Control)

Admin Permissions

Admin can:

- View all market entries

- Create market entries
 - Edit any market entry
 - Delete any market entry
 - Approve or Reject market entries
 - Add/Edit/Delete predefined shopping items
 - Add custom items
 - Upload/Delete receipt images
 - View complete market history
 - Search and Filter market records
 - Lock monthly market records
 - Restore deleted market records
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Member Permissions

Members can:

- Create market entries
- Select items from the predefined shopping list
- Enter quantity and price
- Add custom items
- Upload receipt image
- View their own submitted market records
- View purchased items
- View total market amount

Members cannot

- Edit other members' records
 - Delete other records
 - Delete shopping items
 - Edit shopping items
 - Lock month
 - Restore deleted data
 - Access Backup
 - Access Audit Logs
 - Manage other members' market records
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Market Approval System

Every market entry submitted by a member will have one of the following statuses:

- Pending
- Approved
- Rejected

Only Approved entries will be included in monthly calculations.

Money Collection

Store

- Member
- Amount
- Payment Date
- Payment Method

Payment Methods

- Cash
- bKash
- Nagad
- Bank Transfer

Optional Notes

Expense Management

Categories

- Gas
 - Electricity
 - Water
 - Internet
 - Maid Salary
 - Maintenance
 - Other Expenses
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Monthly Report

Automatically generate

- Total Market Cost
- Total Expenses
- Total Meals
- Meal Rate
- Individual Bill

- Paid Amount
- Due Amount

Support

- Print
 - PDF Export
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Notice Board

Admin

- Create Notice
- Edit Notice
- Delete Notice

Member

- View Notices
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Search & Filter

Search

- Members
- Market
- Payments
- Expenses

Filter

- Date
 - Month
 - Member
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Settings

Allow Admin to configure

- Mess Name
- Address

- Contact Number
 - Default Language
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Multi-Language Support

Support

- English
- Bangla

Requirements

- One-click language switch
 - Remember selected language
 - Translate the complete interface
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Monthly Lock

When monthly accounts are finalized

- No one can edit records
 - Only Admin can unlock the month
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Backup & Restore

Admin can

- Download full database backup
 - Restore backup
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Audit Log

Track

- Login
- Logout
- Member Created
- Member Updated
- Member Deleted

- Market Created
- Market Updated
- Payment Updated
- Expense Updated
- Monthly Lock
- Backup
- Restore

Each log stores

- User
 - Date
 - Time
 - Action
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Member Leave Mode

Allow Admin to set

- Leave Start Date
- Leave End Date

During leave

- Meals are automatically excluded

After leave

- Member becomes active automatically
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Notification Center

Admin receives notifications

- New Member
- New Payment
- New Market Entry
- Monthly Lock
- Backup Completed
- Restore Completed

Member receives notifications

- Payment Recorded

- Due Updated
- New Notice
- Leave Approved
- Monthly Report Finalized

Support

- Read
- Unread

Trash / Recycle Bin

Deleted data should not be permanently removed immediately.

Support restoring

- Members
- Market Entries
- Payments
- Expenses
- Notices

Admin can

- Restore
- Permanently Delete

Automatically delete after 30 days.

Responsive Design

Fully responsive for

- Mobile
- Tablet
- Laptop
- Desktop

Automatic Calculations

Automatically calculate

- Total Market Cost
 - Total Expenses
 - Total Meals
 - Meal Rate
 - Individual Bill
 - Paid Amount
 - Due Amount
 - Remaining Balance
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Admin Panel

- Dashboard
 - Members
 - Meals
 - Market
 - Payments
 - Expenses
 - Reports
 - Notices
 - Settings
 - Notifications
 - Audit Logs
 - Backup & Restore
 - Recycle Bin
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Member Panel

- Dashboard
 - My Meals
 - My Payments
 - My Due
 - My Market Entries
 - Monthly Report
 - Notices
 - Notifications
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User Interface Requirements

- Clean Modern UI
 - Responsive Layout
 - Fast Loading
 - Easy Navigation
 - Mobile First Design
 - Professional Dashboard
 - Cards
 - Tables
 - Charts
 - Search
 - Pagination
 - Confirmation Dialog Before Delete
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Security Requirements

- Password Encryption
 - Secure Authentication
 - Role-Based Access Control (RBAC)
 - Input Validation
 - SQL Injection Protection
 - XSS Protection
 - CSRF Protection
 - Secure API Access
 - Session Management
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Project Goal

The final product should be a secure, modern, scalable, and professional **Mess Management System** suitable for real-world use. It should provide an easy way to manage members, meals, market expenses, payments, and monthly accounting while ensuring data security, transparency, and a smooth user experience on both desktop and mobile devices. The architecture should be future-ready, allowing easy integration of features such as online payments, mobile applications, and multi-mess support.